

安徽大学 2020-2021 学年第一学期《数理方程》期末考 试试卷 (A 卷) 参考答案和评分标准

January 9, 2021

1. 解答题 (30 分)

(1) 解: 由达朗贝尔公式得:

$$u(x, t) = \varphi(x + at) + \varphi(x - at) + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi \quad (1)$$

$$= 0 + \frac{1}{2a} \int_{x-at}^{x+at} a \sin^2 \xi d\xi \quad (2)$$

$$= \frac{at}{2} - \frac{\sin 2(x + at) - \sin 2(x - at)}{8} \quad (3)$$

$$= \frac{at}{2} - \frac{\cos 2x \sin 2at}{4} \quad (4)$$

评分标准: 正确的达朗贝尔公式 2 分, 正确的结果 3 分. 若仅有最终结果, 可得 3 分.

(2) 解: 完整的定解问题如下:

$$u_t - a^2 u_{xx} = 0 \quad (5)$$

这里 $a^2 = \frac{\rho}{ck}$. 边界条件:

$$u_x|_{x=0} = 0, \quad u_x|_{x=l} = 0 \quad (6)$$

初始条件:

$$u|_{t=0} = u_0 x \quad (7)$$

评分标准: 正确的泛定方程 2 分, 正确的边界条件 2 分, 正确的初始条件 1 分.

(3) 解:

$$u_x = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta} \quad (8)$$

$$u_y = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial y} = a \frac{\partial u}{\partial \xi} - a \frac{\partial u}{\partial \eta} \quad (9)$$

进而,

$$u_{xx} = \frac{\partial u_x}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u_x}{\partial \eta} \frac{\partial \eta}{\partial x} \quad (10)$$

$$= \frac{\partial u_x}{\partial \xi} + \frac{\partial u_x}{\partial \eta} \quad (11)$$

$$= \frac{\partial^2 u}{\partial \xi^2} + 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2} \quad (12)$$

$$u_{yy} = \frac{\partial u_y}{\partial \xi} \frac{\partial \xi}{\partial y} + \frac{\partial u_y}{\partial \eta} \frac{\partial \eta}{\partial y} \quad (13)$$

$$= a \frac{\partial u_y}{\partial \xi} - a \frac{\partial u_y}{\partial \eta} \quad (14)$$

$$= a^2 \frac{\partial^2 u}{\partial \xi^2} - 2a^2 \frac{\partial^2 u}{\partial \xi \partial \eta} + a^2 \frac{\partial^2 u}{\partial \eta^2} \quad (15)$$

因此,

$$u_{yy} - a^2 u_{xx} = -4a^2 \frac{\partial^2 u}{\partial \xi \partial \eta} = 0 \quad (16)$$

即原方程化为:

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0 \quad (17)$$

评分标准: 正确的 u_x , u_y , u_{xx} 和 u_{yy} , 各得 1 分, 正确的结果 1 分. 等价的推导过程酌情给分.

(4) 解: 当 $\lambda = 0$ 时, 方程的解可表示为

$$X = Ax + B \quad (18)$$

代入边界条件得:

$$X_x|_{x=0} = A = 0 \quad (19)$$

$$X_x|_{x=l} = A = 0 \quad (20)$$

所以, 当 $\lambda = 0$ 时,

$$X = B \quad (21)$$

当 $\lambda \neq 0$ 时, 方程的解可表示为

$$X = C \cos \sqrt{\lambda} x + D \sin \sqrt{\lambda} x \quad (22)$$

代入边界条件得:

$$X_x|_{x=0} = D\sqrt{\lambda} = 0 \quad (23)$$

故

$$D = 0 \quad (24)$$

从而

$$X_x|_{x=l} = -C\sqrt{\lambda} \sin \sqrt{\lambda} l = 0 \quad (25)$$

于是

$$\sqrt{\lambda}l = k\pi, \quad k = 1, 2, 3, \dots \quad (26)$$

得到本征值:

$$\lambda = \left(\frac{k\pi}{l}\right)^2 \quad (27)$$

和本征函数

$$X_k = \cos \frac{k\pi x}{l}, k = 1, 2, \dots \quad (28)$$

因此, 原本征值问题的本征值为:

$$\lambda = \left(\frac{k\pi}{l}\right)^2, k = 0, 1, 2, \dots \quad (29)$$

相应的本征函数为

$$X_k = \cos \frac{k\pi x}{l}, k = 0, 1, 2, \dots \quad (30)$$

评分标准: $\lambda = 0$ 的情况 1 分, $\lambda \neq 0$ 的情况 4 分.

(5) 解: 由 $x = h$ 处的受力分析可得:

$$T \cos \alpha_1 = T \cos \alpha_2 \quad (31)$$

$$T \sin \alpha_1 + T \sin \alpha_2 = F \quad (32)$$

$F \ll T$, 则 α_1, α_2 是小量,

$$T\alpha_1 + T\alpha_2 = F \quad (33)$$

由图中的几何关系可知,

$$h\alpha_1 = (l - h)\alpha_2 \quad (34)$$

得到

$$\alpha_1 = \frac{F}{Tl} (l - h) \quad (35)$$

$$\alpha_2 = \frac{Fh}{Tl} \quad (36)$$

所以初始位移为:

$$u|_{t=0} = \begin{cases} \frac{F}{Tl} (l - h) x, & 0 \leq x \leq h \\ -\frac{Fh}{Tl} (x - l) & h \leq x \leq l \end{cases} \quad (37)$$

在松手的瞬间, 线上各点来不及加速, 所以各点的初速度为 0:

$$u_t|_{t=0} = 0 \quad (38)$$

评分标准: 正确的初始位移 4 分, 正确的初始速度 1 分.

(6) 原方程可化为:

$$y''(x) + \frac{1}{x}y'(x) + \frac{x^2 - \nu^2}{x^2}y(x) = 0 \quad (39)$$

对于 $x_0 = 0$, 由于

$$\lim_{x \rightarrow x_0} (x - x_0) \frac{1}{x} = 1 \quad (40)$$

和

$$\lim_{x \rightarrow x_0} (x - x_0)^2 \frac{x^2 - \nu^2}{x^2} = -\nu^2 \quad (41)$$

所以, $x_0 = 0$ 是系数 $\frac{1}{x}$ 的一阶奇点, $\frac{x^2 - \nu^2}{x^2}$ 的两阶奇点, 故 $x_0 = 0$ 是原方程的正则奇点.

评分标准: 正确的结论 1 分, 正确的极限各 2 分.

2. 解答题 (15 分)

解 在 $x_0 = 0$ 的邻域上, 设 $y(x) = \sum_{k=0}^{\infty} a_k x^k$, 则

$$y' = \sum_{k=0}^{\infty} a_k k x^{k-1} \quad (42)$$

$$y'' = \sum_{k=0}^{\infty} a_k k(k-1) x^{k-2} \quad (43)$$

于是,

$$x^2 y'' = \sum_{k=0}^{\infty} a_k k(k-1) x^k \quad (44)$$

$$y'' = \sum_{k=0}^{\infty} a_{k+2} (k+2)(k+1) x^k \quad (45)$$

$$xy' = \sum_{k=0}^{\infty} a_k k x^k \quad (46)$$

..... 5 分

代入原方程得:

$$\sum_{k=0}^{\infty} a_k k(k-1) x^k - \sum_{k=0}^{\infty} a_{k+2} (k+2)(k+1) x^k + \sum_{k=0}^{\infty} a_k k x^k - \sum_{k=0}^{\infty} a_k x^k = 0 \quad (47)$$

即

$$\sum_{k=0}^{\infty} (a_k k(k-1) - a_{k+2} (k+2)(k+1) + a_k k - a_k) x^k = 0 \quad (48)$$

由泰勒级数的唯一性,

$$a_k k(k-1) - a_{k+2} (k+2)(k+1) + a_k k - a_k = 0 \quad (49)$$

$$a_{k+2} = \frac{k-1}{k+2} a_k \quad (50)$$

所以, 奇次项和偶次项分别构成两个线性独立的解.

..... 4 分.....

当 k 为偶数时,

$$a_{2n} = \frac{2n-2-1}{2n-2+2} a_{2n-2} \quad (51)$$

$$= \frac{2n-3}{2n} \frac{2n-4-1}{2n-4+2} a_{2n-4} \quad (52)$$

$$= \frac{2n-3}{2n} \frac{2n-5}{2n-2} a_{2n-4} \quad (53)$$

$$= \dots \quad (54)$$

$$= \frac{2n-3}{2n} \frac{2n-5}{2n-2} \frac{2n-7}{2n-4} \dots \frac{-1}{2} a_0 \quad (55)$$

$$= \frac{\prod_{i=1}^n (2i-3)}{2^n (n!)} a_0 \quad (56)$$

所以

$$y_0(x) = a_0 + a_0 \sum_{n=1}^{\infty} \frac{\prod_{i=1}^n (2i-3)}{2^n (n!)} x^{2n} \quad (57)$$

..... 3 分.....

当 k 为奇数时,

$$a_3 = 0 \quad (58)$$

所以 $a_{2n+1} = 0$ ($n = 1, 2, 3, \dots$), 于是

$$y_1(x) = a_1 x \quad (59)$$

..... 3 分.....

3 解答题 (15 分)

解 (1) 由已知,

$$P_{l+1} = \frac{2l+1}{l+1} x P_l - \frac{l}{l+1} P_{l-1} \quad (60)$$

已知 $P_0 = 1, P_1 = x$, 则

$$P_2 = \frac{3}{2} x P_1 - \frac{1}{2} P_0 = \frac{3x^2}{2} - \frac{1}{2} \quad (61)$$

$$P_3 = \frac{5}{3} x P_2 - \frac{2}{3} P_1 = \frac{5}{3} x \left(\frac{3x^2}{2} - \frac{1}{2} \right) - \frac{2}{3} x = \frac{5}{2} x^3 - \frac{3}{2} x \quad (62)$$

评分标准: 正确的 P_2 和 P_3 分别得 1 分和 2 分.

(2) 令 $f(x) = c_0 P_0 + c_1 P_1 + c_2 P_2 + c_3 P_3$,

..... 3 分.....

则

$$f(x) = c_0 + c_1 x + c_2 \left(\frac{3x^2}{2} - \frac{1}{2} \right) + c_3 \left(\frac{5}{2} x^3 - \frac{3}{2} x \right) \quad (63)$$

$$= c_0 - \frac{c_2}{2} + \left(c_1 - \frac{3}{2} c_3 \right) x + \frac{3c_2}{2} x^2 + \frac{5c_3}{2} x^3 \quad (64)$$

已知 $f(x) = a_3x^3 + a_2x^2 + a_1x + a_0$, 所以

$$\begin{cases} c_0 - \frac{c_2}{2} &= a_0 \\ c_1 - \frac{3}{2}c_3 &= a_1 \\ \frac{3c_2}{2} &= a_2 \\ \frac{5c_3}{2} &= a_3 \end{cases} \quad (65)$$

..... **3 分**

解得

$$\begin{cases} c_0 &= a_0 + \frac{1}{3}a_2 \\ c_1 &= a_1 + \frac{3}{5}a_3 \\ c_2 &= \frac{2}{3}a_2 \\ c_3 &= \frac{2}{5}a_3 \end{cases} \quad (66)$$

所以

$$f(x) = \left(a_0 + \frac{1}{3}a_2\right)P_0 + \left(a_1 + \frac{3}{5}a_3\right)P_1 + \frac{2}{3}a_2P_2 + \frac{2}{5}a_3P_3 \quad (67)$$

..... **2 分**

评分标准: 等价的正确做法酌情给分.

(3) 积分

$$\int_{-1}^1 P_1(x) P_3(x) dx = \int_{-1}^1 x \left(\frac{5}{2}x^3 - \frac{3}{2}x \right) dx \quad (68)$$

$$= \frac{5}{2} \int_{-1}^1 x^4 dx - \frac{3}{2} \int_{-1}^1 x^2 dx \quad (69)$$

$$= \frac{5}{2} \frac{x^5}{5} \Big|_{-1}^1 - \frac{3}{2} \frac{x^3}{3} \Big|_{-1}^1 \quad (70)$$

$$= \frac{5}{2} \frac{2}{5} - \frac{3}{2} \frac{2}{3} \quad (71)$$

$$= 0 \quad (72)$$

$$\int_{-1}^1 P_2^2(x) dx = \int_{-1}^1 \left(\frac{3x^2}{2} - \frac{1}{2} \right)^2 dx \quad (73)$$

$$= \int_{-1}^1 \left(\frac{9x^4}{4} - \frac{3}{2}x^2 + \frac{1}{4} \right) dx \quad (74)$$

$$= \frac{9}{4} \frac{x^5}{5} \Big|_{-1}^1 - \frac{3}{2} \frac{x^3}{3} \Big|_{-1}^1 + \frac{1}{4} x \Big|_{-1}^1 \quad (75)$$

$$= \frac{9}{4} \frac{2}{5} - \frac{3}{2} \frac{2}{3} + \frac{1}{2} \quad (76)$$

$$= \frac{2}{5} \quad (77)$$

评分标准: 每个积分 2 分.

4 解答题 (20 分)

解 边界条件确定的完备本征函数为 $X_n = \sin\left(\frac{n\pi}{l}x\right)$, 本征值 $\lambda_n = \frac{n^2\pi^2}{l^2}$, $n = 1, 2, \dots$. 则 u 的一般解为:

$$u = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l} \quad (78)$$

..... 4 分

代入泛定方程得

$$\sum_{n=1}^{\infty} T_n''(t) \sin \frac{n\pi x}{l} - a^2 \sum_{n=1}^{\infty} T_n(t) \frac{-n^2\pi^2}{l^2} \sin \frac{n\pi x}{l} = 0 \quad (79)$$

即

$$\sum_{n=1}^{\infty} \left(T_n''(t) + \frac{n^2\pi^2 a^2}{l^2} T_n(t) \right) \sin \frac{n\pi x}{l} = 0 \quad (80)$$

所以

$$T_n''(t) + \frac{n^2\pi^2 a^2}{l^2} T_n(t) = 0 \quad (81)$$

它的解可写为

$$T_n = A_n \cos \frac{n\pi a t}{l} + B_n \sin \frac{n\pi a t}{l} \quad (82)$$

..... 6 分

于是, 方程的解可写为

$$u = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a t}{l} + B_n \sin \frac{n\pi a t}{l} \right) \sin \frac{n\pi x}{l} \quad (83)$$

..... 3 分

代入初始条件得:

$$u|_{t=0} = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = u_0 \sin \frac{3\pi x}{l} \quad (84)$$

$$u_t|_{t=0} = \sum_{n=1}^{\infty} B_n \frac{n\pi a}{l} \sin \frac{n\pi x}{l} = \psi(x) \quad (85)$$

上述两式, 两边同乘以 $\sin \frac{m\pi x}{l}$, 并对 x 在区间 $[0, l]$ 上积分; 利用 $\int_0^l \sin \frac{m\pi x}{l} \sin \frac{n\pi x}{l} dx = \frac{l}{2} \delta_{mn}$ 得:

$$\sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} \frac{l}{2} \delta_{mn} = u_0 \int_0^l \sin \frac{3\pi x}{l} \sin \frac{m\pi x}{l} dx \quad (86)$$

$$\sum_{n=1}^{\infty} B_n \frac{n\pi a}{l} \sin \frac{n\pi x}{l} \frac{l}{2} \delta_{mn} = \int_0^l \psi(x) \sin \frac{m\pi x}{l} dx \quad (87)$$

那么

$$\frac{l}{2} A_m = u_0 \int_0^l \sin \frac{3\pi x}{l} \sin \frac{m\pi x}{l} dx \quad (88)$$

$$B_m \frac{m\pi a}{l} \frac{l}{2} = \int_0^l \psi(x) \sin \frac{m\pi x}{l} dx \quad (89)$$

即

$$A_m = u_0 \delta_{m3} \quad (90)$$

$$B_m = \frac{2}{m\pi a} \int_0^l \varphi(x) \sin \frac{m\pi x}{l} dx \quad (91)$$

..... 6 分

总结, 定解问题的解为

$$u = u_0 \cos \frac{3\pi at}{l} \sin \frac{3\pi x}{l} + \sum_{n=1}^{\infty} \frac{2}{n\pi a} \int_0^l \psi(\xi) \sin \frac{n\pi \xi}{l} d\xi \sin \frac{n\pi at}{l} \sin \frac{n\pi x}{l} \quad (92)$$

..... 1 分

评分标准: 若按照标准的分离变量法求解: 给出解的形式 $u = X(t)T(t)$ (1 分); 分离出 X 和 T 的常微分方程和边界条件 (4 分); X 的本征值问题的正确解 (4 分); T 的求解 (4 分); 正确的一般解 (2 分); 代入初始条件, 并求出正确的系数 (4 分); 正确的结果形式 (1 分).

5 解答题 (20 分)

解 (1) 把 $u(x, t) = w(x, t) + v(x)$ 代入定解问题得:

$$w_t - a^2 w_{xx} = \sin \frac{\pi x}{2l} - v_t + a^2 v_{xx}, \quad (93)$$

$$w|_{x=0} = u_0 - v|_{x=0}, \quad (94)$$

$$w_x|_{x=l} = \frac{u_0}{l} - v_x|_{x=l}, \quad (95)$$

$$w|_{t=0} = \frac{2u_0 x}{l} + \left(\frac{l}{a\pi}\right)^2 \sin \frac{\pi x}{2l} - v|_{t=0} \quad (96)$$

..... 2 分

由于

$$v|_{x=0} = u_0 \quad (97)$$

$$v_x|_{x=l} = \frac{u_0}{l} \quad (98)$$

$$v|_{t=0} = u_0 + \frac{u_0 x}{l} + \left(\frac{2l}{a\pi}\right)^2 \sin \frac{\pi x}{2l} \quad (99)$$

..... 1 分

所以,

$$w_t - a^2 w_{xx} = 0 \quad (100)$$

$$w|_{x=0} = 0 \quad (101)$$

$$w_x|_{x=l} = 0 \quad (102)$$

$$w|_{t=0} = -u_0 + \frac{u_0 x}{l} - \frac{3l^2}{a^2 \pi^2} \sin \frac{\pi x}{2l} \quad (103)$$

..... 1 分

(2) 边界条件确定的完备本征函数为 $X_n = \sin\left(\frac{n+\frac{1}{2}}{l}\pi x\right)$, 本征值 $\lambda_n = \frac{(n+\frac{1}{2})^2\pi^2}{l^2}$, $n = 0, 1, 2, \dots$. 则 w 的一般解为:

$$w = \sum_{n=0}^{\infty} T_n(t) \sin\left(\frac{n+\frac{1}{2}}{l}\pi x\right) \quad (104)$$

..... 4 分

代入泛定方程得

$$\sum_{n=0}^{\infty} T'_n(t) \sin\left(\frac{n+\frac{1}{2}}{l}\pi x\right) - a^2 \sum_{n=0}^{\infty} T_n(t) \frac{(n+\frac{1}{2})^2\pi^2}{l^2} \sin\left(\frac{n+\frac{1}{2}}{l}\pi x\right) = 0 \quad (105)$$

即

$$\sum_{n=0}^{\infty} \left(T'_n(t) + \frac{a^2 (n+\frac{1}{2})^2 \pi^2}{l^2} T_n(t) \right) \sin\left(\frac{n+\frac{1}{2}}{l}\pi x\right) = 0 \quad (106)$$

所以

$$T'_n(t) + \frac{(n+\frac{1}{2})^2 \pi^2 a^2}{l^2} T_n(t) = 0 \quad (107)$$

它的解为

$$T_n = C_n \exp\left[-\frac{(n+\frac{1}{2})^2 \pi^2 a^2}{l^2} t\right], \quad n = 0, 1, 2, \dots \quad (108)$$

..... 5 分

于是

$$w(x, t) = \sum_{n=0}^{\infty} C_n \exp\left[-\frac{(n+\frac{1}{2})^2 \pi^2 a^2}{l^2} t\right] \sin\left(\frac{n+\frac{1}{2}}{l}\pi x\right) \quad (109)$$

代入初始条件得:

$$\sum_{n=0}^{\infty} C_n \sin\left(\frac{n+\frac{1}{2}}{l}\pi x\right) = -u_0 + \frac{u_0 x}{l} - \frac{3l^2}{a^2 \pi^2} \sin \frac{\pi x}{2l} \quad (110)$$

两边同时乘以 $\sin\left(\frac{m+\frac{1}{2}}{l}\pi x\right)$, 并对 x 在区间 $[0, l]$ 积分得:

$$\sum_{n=0}^{\infty} C_n \frac{l}{2} \delta_{mn} = \int_0^l \left(-u_0 + \frac{u_0 x}{l} - \frac{3l^2}{a^2 \pi^2} \sin \frac{\pi x}{2l} \right) \sin\left(\frac{m+\frac{1}{2}}{l}\pi x\right) dx, \quad (111)$$

$$\int_0^l \sin\left(\frac{m+\frac{1}{2}}{l}\pi x\right) dx = -\frac{\cos \frac{m+\frac{1}{2}}{l}\pi x}{\frac{m+\frac{1}{2}}{l}\pi} \Big|_0^l = \frac{l}{(m+\frac{1}{2})\pi} \quad (112)$$

$$\int_0^l x \sin\left(\frac{m+\frac{1}{2}}{l}\pi x\right) dx = -x \frac{\cos \frac{m+\frac{1}{2}}{l}\pi x}{\frac{m+\frac{1}{2}}{l}\pi} \Big|_0^l + \int_0^l \frac{\cos \frac{m+\frac{1}{2}}{l}\pi x}{\frac{m+\frac{1}{2}}{l}\pi} dx \quad (113)$$

$$= \frac{\sin \frac{m+\frac{1}{2}}{l}\pi x}{\left(\frac{m+\frac{1}{2}}{l}\pi\right)^2} \Big|_0^l \quad (114)$$

$$= \frac{(-1)^m}{\left(\frac{m+\frac{1}{2}}{l}\pi\right)^2} \quad (115)$$

$$\int_0^l \sin \frac{\pi x}{2l} \sin \left(\frac{m + \frac{1}{2}}{l} \pi x \right) dx = \frac{l}{2} \delta_{m0} \quad (116)$$

所以

$$\frac{l}{2} C_m = -\frac{u_0 l}{\left(m + \frac{1}{2}\right) \pi} + \frac{u_0 l (-1)^m}{\left(m + \frac{1}{2}\right)^2 \pi^2} - \frac{3l^2}{a^2 \pi^2} \frac{l}{2} \delta_{m0} \quad (117)$$

即

$$C_m = -\frac{2u_0}{\left(m + \frac{1}{2}\right) \pi} + \frac{2u_0 (-1)^m}{\left(m + \frac{1}{2}\right)^2 \pi^2} - \frac{3l^2}{a^2 \pi^2} \delta_{m0} \quad (118)$$

..... **6 分**

总结, 定解问题的解为

$$u = u_0 + \frac{u_0 x}{l} + \left(\frac{2l}{a\pi} \right)^2 \sin \frac{\pi x}{2l} - \frac{3l^2}{a^2 \pi^2} \exp \left[-\frac{\pi^2 a^2}{4l^2} t \right] \sin \frac{\pi x}{2l} \quad (119)$$

$$+ \frac{2u_0}{\pi} \sum_{n=0}^{\infty} \left(-\frac{1}{n + \frac{1}{2}} + \frac{(-1)^n}{\left(n + \frac{1}{2}\right)^2 \pi^2} \right) \exp \left[-\frac{\left(n + \frac{1}{2}\right)^2 \pi^2 a^2}{l^2} t \right] \sin \left(\frac{n + \frac{1}{2}}{l} \pi x \right) \quad (120)$$

..... **1 分**

评分标准: 若按照标准的分离变量法求解: 给出解的形式 $u = X(t)T(t)$ (**1 分**); 分离出 X 和 T 的常微分方程和边界条件 (**3 分**); X 的本征值问题的正确解 (**3 分**); T 的求解 (**2 分**); 正确的一般解 (**2 分**); 代入初始条件, 并求出正确的系数 (**4 分**); 正确的结果形式 (**1 分**).